

Def. Let (X, d) be a Metric space.

A sequence is said to be Convergent if: for some $x \in X$, for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t.
 $n \geq N \Rightarrow d(x_n, x) < \epsilon$.

Lemma. Let $\{x_n\}, \{y_n\}$ be Convergent sequences in $\mathbb{R}$.

$$1. \left(\lim_{n \rightarrow \infty} x_n \right) + \left(\lim_{n \rightarrow \infty} y_n \right) = \lim_{n \rightarrow \infty} (x_n + y_n)$$

$$2. \alpha \cdot \left(\lim_{n \rightarrow \infty} x_n \right) = \lim_{n \rightarrow \infty} (\alpha x_n).$$

$$3. \left(\lim_{n \rightarrow \infty} x_n \right) \cdot \left(\lim_{n \rightarrow \infty} y_n \right) = \lim_{n \rightarrow \infty} (x_n \cdot y_n)$$

$$4. \text{ If } \lim_{n \rightarrow \infty} y_n \neq 0 \text{ and } y_i \neq 0 \forall i \in \mathbb{N}, \text{ then } \left(\lim_{n \rightarrow \infty} y_n \right)^{-1} = \lim_{n \rightarrow \infty} y_n^{-1}$$

Proof. Denote $x, y \in X$ are limits of $\{x_n\}, \{y_n\}$, respectively.

Let $\epsilon > 0$ be given. Then, there exists $N_1, N_2 \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow |x_n - x| < \frac{\epsilon}{2} \quad \text{and} \quad n \geq N_2 \Rightarrow |y_n - y| < \frac{\epsilon}{2}$$

By the triangle inequality,

$$n \geq \max\{N_1, N_2\} \Rightarrow |x_n + y_n - (x + y)| \leq |x_n - x| + |y_n - y| < \epsilon.$$

Let $\epsilon > 0$ be given. Then, there exists $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow |x_n - x| < \frac{\epsilon}{|\alpha|}$$

$$\Rightarrow |\alpha| |x_n - x| = |\alpha x_n - \alpha x| < \epsilon. \quad (\text{If } \alpha = 0, \text{ trivial. Thus let } \alpha \neq 0)$$

Let $\epsilon > 0$ be given. Then, there exists $N_1, N_2 \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow |x_n - x| < \sqrt{\epsilon} \quad \text{and} \quad n \geq N_2 \Rightarrow |y_n - y| < \sqrt{\epsilon}.$$

Then, for $N = \max\{N_1, N_2\}$,

$$n \geq N \Rightarrow |(x_n - x)(y_n - y)| < \epsilon$$

Therefore, $\lim_{n \rightarrow \infty} (x_n - x)(y_n - y) = 0$. Now, for all $n \in \mathbb{N}$,

$$0 \leq |x_n y_n - xy| = |x_n - x| \cdot |y_n - y| + x |y_n - y| + y |x_n - x| \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

Thus $x_n y_n \rightarrow xy$.

Let $\epsilon > 0$ be given. Choose $N_1 \in \mathbb{N}$ s.t. $n \geq N_1 \Rightarrow |y_n - y| < \frac{1}{2} |y|$. Then, $n \geq N_1 \Rightarrow \frac{|y|}{2} < |y_n|$.

Meanwhile, choose $N_2 \in \mathbb{N}$ s.t. $n \geq N_2 \Rightarrow |y_n - y| < \frac{1}{2} |y|^2 \epsilon$. Now, for $n \geq \max\{N_1, N_2\}$,

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \left| \frac{y_n - y}{y_n y} \right| < \frac{2}{|y|^2} |y_n - y| \cdot \epsilon = \epsilon.$$

Thm. Every Convergent sequence in a Metric space is Cauchy sequence.

Proof. Let $\{x_n\}$ be a Convergent sequence in a Metric space (X, d) , which converges to $x \in X$.

Let $\varepsilon > 0$ be given. Choose $N \in \mathbb{N}$ s.t

$$n \geq N \Rightarrow d(x_n, x) < \frac{\varepsilon}{2}$$

$$d(a_m, a_n) \leq d(a_m, a_N) + d(a_N, a_n)$$

Then, by the triangle inequality,

$$m, n \geq N \Rightarrow d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon.$$

Thm. Every Cauchy sequence in a Compact Metric space is Convergent.

Proof. Let $\{a_n\}$ be a Cauchy sequence in a Compact Metric space (X, d) .

Consider a map diameter of subset in Metric space

$$\text{diam} E: \mathcal{P}(X) \rightarrow \mathbb{R}; E \mapsto \sup \{d(p, q) \mid p, q \in E\}$$

This map satisfies: $\text{diam} E = \text{diam} \bar{E}$, for all $E \subseteq X$, because: $\text{diam} \bar{E} \leq \text{diam} E$ is clear, let $\varepsilon > 0$ be given. Then, for any $p, q \in \bar{E}$, there exists $p', q' \in E$ s.t

$$d(p, p') < \frac{\varepsilon}{2} \quad \text{and} \quad d(q, q') < \frac{\varepsilon}{2}$$

Now, the triangle inequality gives

$$d(p', q') \leq d(p', p) + d(p, q) + d(q, q') < \varepsilon + \text{diam} E.$$

Therefore $\text{diam} \bar{E} \leq \text{diam} E$, $\text{diam} \bar{E} = \text{diam} E$.

Take $E_n = \{a_i \mid i \geq n\}$. By definition of Cauchy sequence, $\lim_{n \rightarrow \infty} \text{diam} E_n = 0$.

Moreover, Combining above results

$$\lim_{n \rightarrow \infty} \text{diam} \bar{E}_n = \lim_{n \rightarrow \infty} \text{diam} E_n = 0.$$

Since Closed set in Compact Space is Compact, and by definition, $\{\bar{E}_n\}$ is Compact sets s.t

$$\bar{E}_1 \supset \bar{E}_2 \supset \dots \quad \bar{E}_i \neq \emptyset.$$

Then $\bigcap_{n=1}^{\infty} \bar{E}_n$ is a singleton, because: if $\bigcap_{n=1}^{\infty} \bar{E}_n$ is empty, then

$$\bar{E}_1 \cap \bigcap_{n=2}^{\infty} \bar{E}_n = \emptyset \quad \text{which implies} \quad \bar{E}_1 \subseteq \bigcup_{n=2}^{\infty} \bar{E}_n^c$$

Since $\bar{E}_1$ is Compact, for some finite covers, $2 \leq n_1 < \dots < n_r$,

$$\bar{E}_1 \subseteq \bar{E}_{n_1}^c \cup \dots \cup \bar{E}_{n_r}^c \quad \text{which implies} \quad \bar{E}_1 \cap \bar{E}_{n_1} \cap \dots \cap \bar{E}_{n_r} = \bar{E}_{n_r} = \emptyset.$$

Contradiction. And if it contains more than one point, then $\lim_{n \rightarrow \infty} \text{diam} \bar{E}_n > 0$, Contradiction

Finally, let ε_0 be given. Then, there exists $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow \text{diam } \overline{E}_n < \varepsilon.$$

Now, for $a \in \bigcap_{n=1}^{\infty} \overline{E}_n$,

$$\forall N \Rightarrow d(a_k, a) \leq \text{diam } \overline{E}_k < \varepsilon.$$

Corollary, $\mathbb{R}^n$ is Complete Metric Space.

Proof. Every Cauchy sequence in a Metric space is bounded,
we can cover this by compact subset of $\mathbb{R}^n$.

Special Sequence in $\mathbb{R}$.

Thm.

1) If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.

2) If $p > 0$, then $\lim_{n \rightarrow \infty} \sqrt[n]{p} = 1$

3). $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

4) If $\alpha \in \mathbb{R}$ and $x > 1$, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{x^n} = 0$.

Proof Let $\epsilon > 0$ be given.

1) Choose $N = \left(\frac{1}{\epsilon}\right)^{\frac{1}{p}}$. Then,

$$x = 1 + p$$

$$n \geq N \Leftrightarrow n \geq \left(\frac{1}{\epsilon}\right)^{\frac{1}{p}} \Rightarrow \frac{1}{n^p} < \epsilon$$

2) If $p = 1$, trivial. If $p > 1$, take $x_n = \sqrt[p]{p} - 1$. The Binomial Theorem gives

$$p = (x_n + 1)^n = \sum_{k=0}^n \binom{n}{k} x_n^k \geq 1 + n \cdot x_n \text{ which implies } x_n \leq \frac{p-1}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

If $p < 1$, $\lim_{n \rightarrow \infty} \sqrt[n]{p} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{\frac{1}{p}}} = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{p}}}$ converges.

3). Take $x_n = \sqrt[n]{n} - 1$. The Binomial Thm gives

$$n = (x_n + 1)^n = \sum_{k=0}^n \binom{n}{k} x_n^k \geq \frac{n(n-1)}{2} x_n^2 \text{ which implies } x_n \leq \sqrt{\frac{2}{n-1}} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

4). Take $k \in \mathbb{N}$ s.t. $k \geq \alpha$. Then, for all $n \geq 2k$,

$$x^n = (1 + (x-1))^n \geq \binom{n}{k} (x-1)^k = \frac{n(n-1)(n-2) \cdots (n-k+1)}{k!} \cdot (x-1)^k \geq \frac{n^k}{2^k} \cdot \frac{(x-1)^k}{k!}$$

which implies that

$$\frac{n^\alpha}{x^n} \leq n^{\alpha-k} \cdot \frac{2^k k!}{(x-1)^k} \rightarrow 0 \text{ as } n \rightarrow \infty.$$